



Recommendation System Project

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[Live Demo](#)



Project Summary

Focus:

- Building a Recommendation System that delivers personalised suggestions based on historical user data, such as past interactions, preferences, and behaviours.

Aim:

- to enhance user satisfaction, engagement, and retention by offering relevant product.



Business Overview & Questions

- Task 1: Recommender System – Hybrid (Rule-based + Logistic Regression)
- Task 2: Anomaly Detection – Isolation Forest
 - Streamlit app with interactive tabs
 - Improved accuracy and fraud detection

01

What categories should we recommend next for a user given their recent behaviour?

02

Which users exhibit abnormal behaviour (e.g., bots/bulk buyers) that should be excluded from modelling and analytics?

03

How much does data cleaning/anomaly removal improve recommendation quality?



Dataset Overview

01

Events Dataset

Timestamp
Visitor_id
Event (view, add to cart,
transaction)
Item_id
Transactionid

02

Category Dataset

Categoryid
Parent_id

03

Item Properties 1 & 2

Timestamp
Item_id
Property
Value

04

Sample Dataset

Input columns:
timestamp, visitorid,
event, categoryid_final
Sample: 30k rows, 200
users



01.**Data Ingestion & Cleaning**

Load raw events data
Standardise formats
Exploratory data analysis
Removed outliers.

02.**Feature Engineering**

Construct candidate features
Generate SVD embeddings for
users and items to capture
latent preferences.

03.**Model Training**

Train baseline models (Logistic
Regression, Random Forest,
Gradient Boosting).
Build Hybrid Recommender
combining baseline features +
SVD embeddings.
Train anomaly detection model to
identify abnormal user behaviour.

System Flow Diagram

04.**Evaluation**

Apply precision, recall, F1 for
anomaly detection.
Assess recommender with
coverage, top-k hit rate, and
accuracy.
Compare "before vs after
cleaning" to validate
improvements.

05.**App Integration(Streamlit)**

Develop interactive UI with two
key modules:
Recommendations (Top-K
personalised categories).
Anomaly Audit (flagging
suspicious users).Add debug
panels for user activity insights.

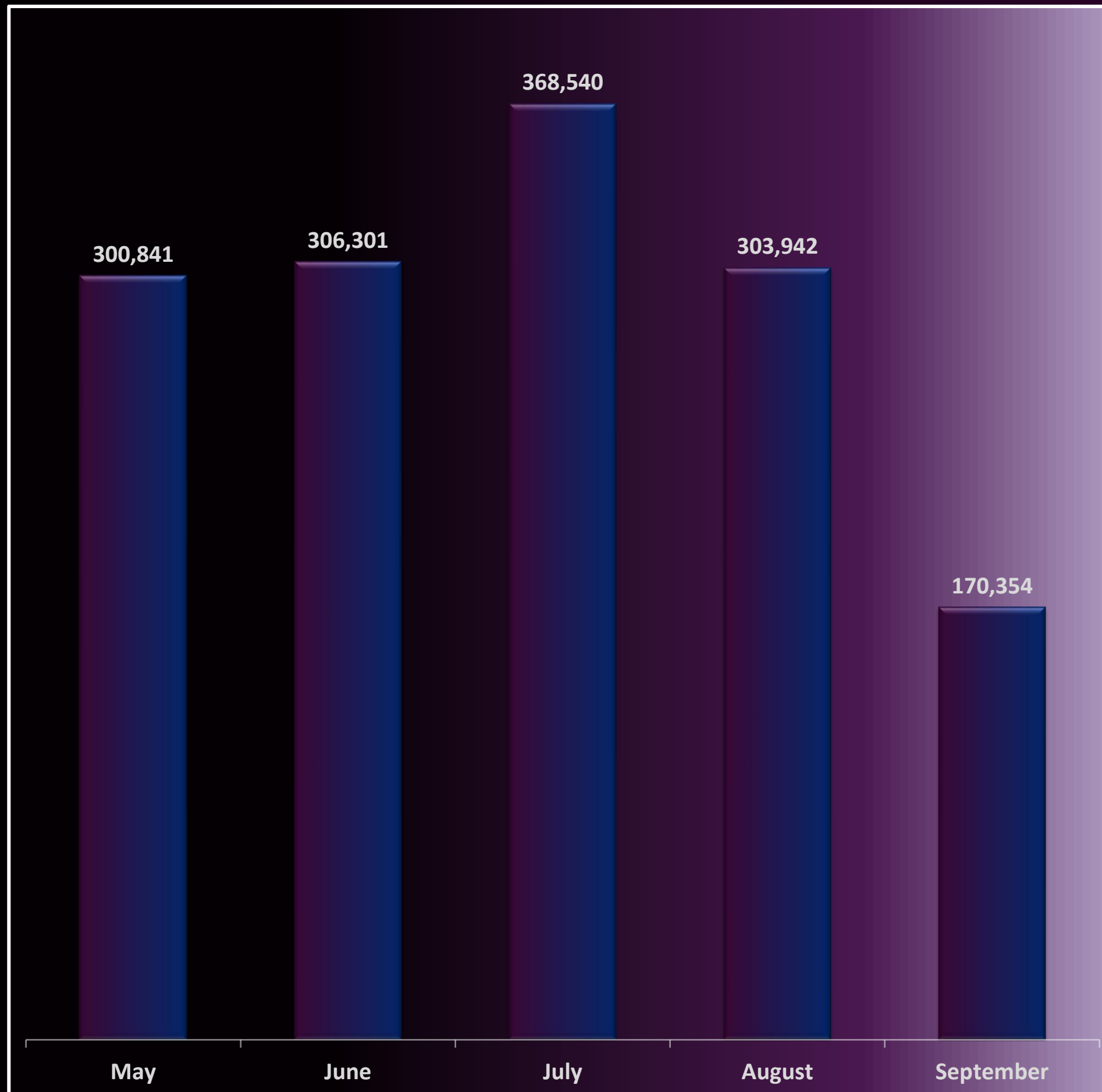
06.**Deployment**

Package artefacts (.pkl,
embeddings, metadata, sample
data).
Deploy Streamlit app on
Streamlit Cloud.



EXPLORATORY DATA ANALYSIS





01.

Seasonality and Retention

- User activity peaks in July (368K), while September drops sharply (170K).
- There is a seasonal effect, possibly linked to summer promotions, holidays, or marketing campaigns in July.
- Business could allocate higher budgets to peak seasons and design re-engagement strategies for off-peak months like September.

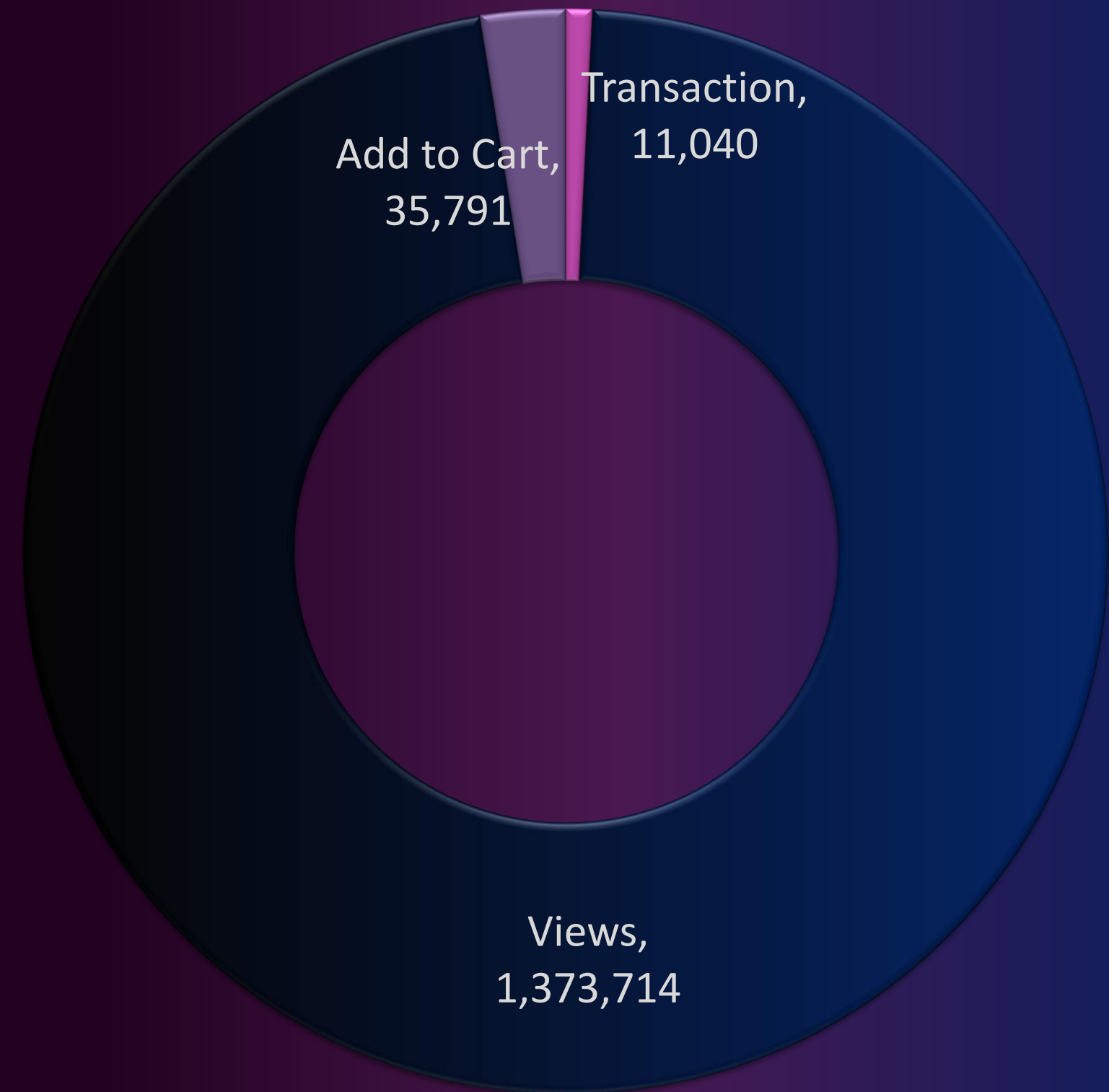
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02.

E-commerce Conversion

- Out of 1.37M users who viewed products, only 35.7K added to cart, and just 11K completed a transaction.
- There is a steep drop-off at each stage – less than 1% of viewers actually purchase.
- This signals big opportunities to improve product appeal, checkout flow, or trust in the platform.

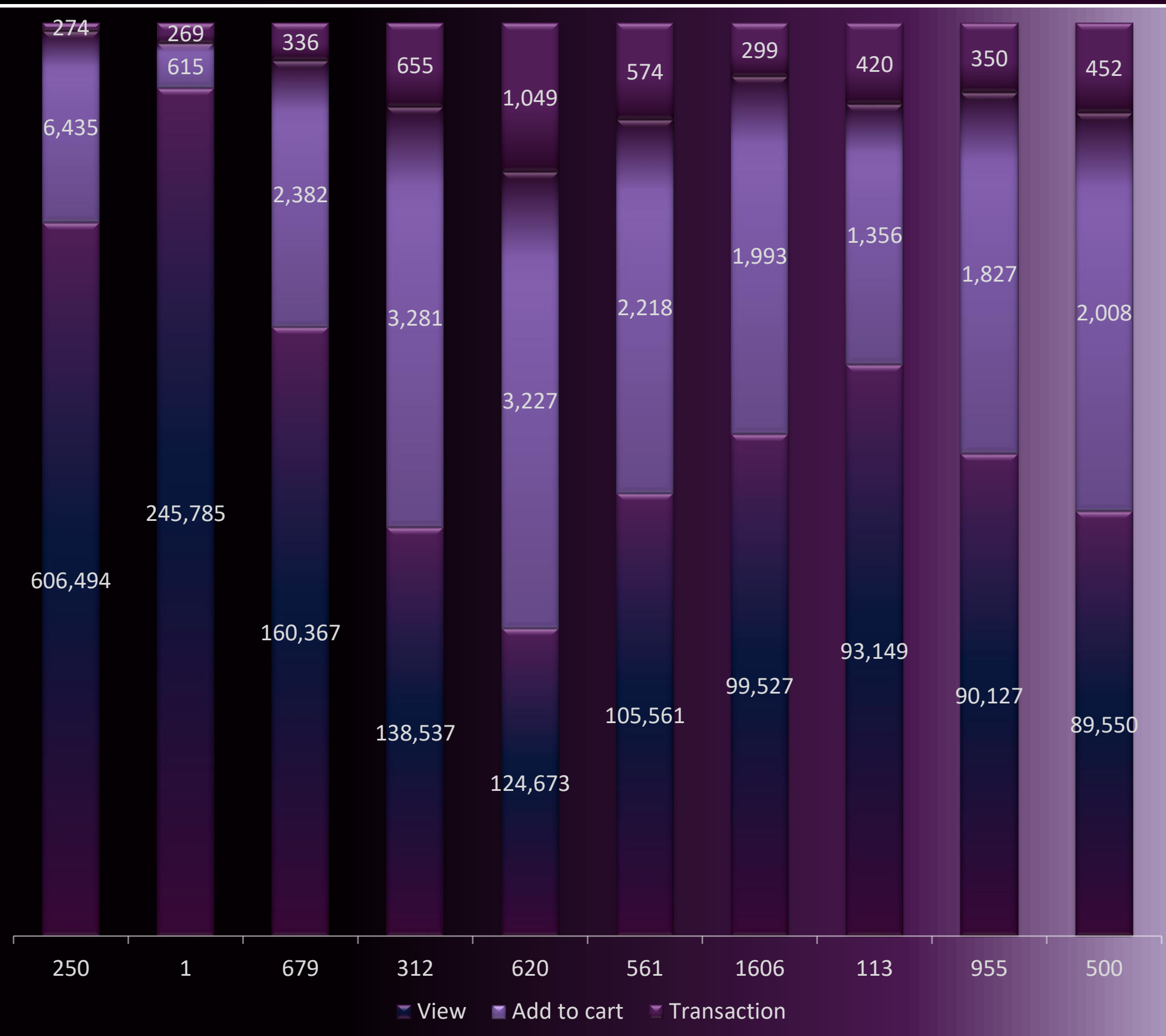


03.

Category Popularity

- Observation: Category 250 dominates with over 6M interactions (606K views alone). Other strong categories include 679, 312, 620, and 561.
- User attention is highly skewed toward a few categories.
- These categories should be prioritised in recommendations, inventory management, and promotions.

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MACHINE LEARNING

Feature Engineering

- Preprocessing:
 - Normalized timestamps
 - Dropped invalid rows
 - Sharded embeddings
- Engineered user-level features for anomaly detection

Model Training

Task 1 – Hybrid Recommender:

- Logistic Regression, RandomForest, HistGradient, SVD, Hybrid SVD.

Task 2 – Anomaly Detection:

- Isolation Forest

Task 1 Model Evaluation

- Hybrid SVD (Champion Model)
- Combined handcrafted baseline features + SVD embeddings into a hybrid feature set
- Performance: Coverage ~91.6%, Top1 ~85.6%, Top3 ~99.5%. Selected as production champion.

Task 2 Model Evaluation

- Proxy anomaly rate of ~2.08%
- Model trained on engineered user behaviour features.
- Chosen contamination: 0.05 → balances recall and precision.
- Performance: Precision = 31%, Recall = 75%, F1 = 0.44.
- Flagged share: 5% of users identified as potentially anomalous.





Streamlit App Deployment



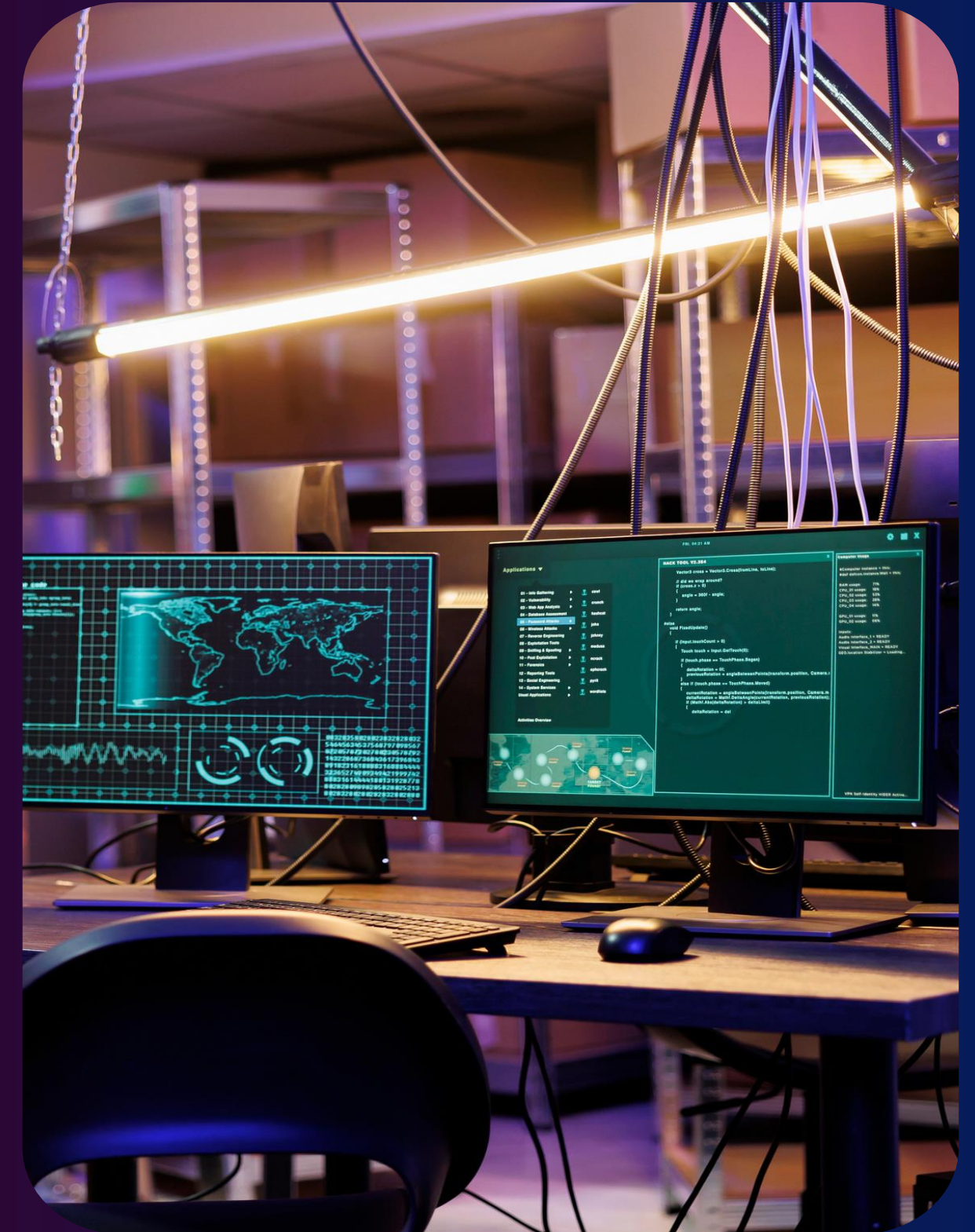
Recommendations

- Upload dataset or use sample
- Pick user, adjust parameters
- Exclude anomalous users



Anomaly Audit

- Load user features
- Run Isolation Forest
- Download flagged users



Business

Implications



- Early detection of bots/bulk accounts → protects GMV analytics and inventory signals
- Category recommendations across homepage, product page, and email
- Fraud/abuse mitigation and traffic quality monitoring





- Candidate-label coupling: ensure strict temporal splits & leakage checks.
- Calibrate anomaly threshold to business tolerance; review flagged cohorts regularly.
- Cold-start: add content-based features and back-off rules.
- Monitor drift; schedule retraining (e.g., weekly).

Risks & Considerations



Appendix

Links

- Recommender live Demo Link : <https://recommendation-system-project-c8cgulvvsynvm77sfyfwb.streamlit.app/>
- Github Link : <https://github.com/julietkukuia/Recommendation-System-Project/tree/main>



Thank You!

A recommendation system is only as good as the trust it builds.
By combining predictive accuracy with anomaly detection, we
not only improve user experience but also safeguard the
integrity of the platform.